

Core Business & Security Rules

Submission Note: Testing credentials for all roles (Admin, Manager, Member) are provided in the attached Credentials Excel Sheet (attached in the submission zip).

Restricted Admin Access: Administrative accounts are pre-configured and cannot be created via the public registration form.

Role-Based Permissions:

Admins Only: Delete projects; Enable/Disable user accounts.

Managers Only: Create new projects.

Data Integrity: All associated tasks are automatically deleted when a project is removed (Cascading Delete).

Security Protocol: * For security, Admin passwords cannot be reset via the public email tool.

Features

User Management

- User registration and login
- Role-based access (Admin, Manager, Member)
- Password reset via email
- Profile management
- Passwords stored using Django's encrypted hashing
- CSRF protection enabled for all AJAX requests

Project & Task Management

- Managers can create projects and create and assign tasks
- Tasks include status, priority, and due date
- Members see assigned tasks on their dashboard and update as needed
- Admins can delete projects (with cascading task deletion)

Messaging System

- One-to-one messaging between users
- Recent chats list
- AJAX-based message sending
- Polling every 30 seconds for new messages

Admin Panel

- System-wide dashboard with statistics
- View and manage all projects

- Enable/disable users
- Full oversight of system data

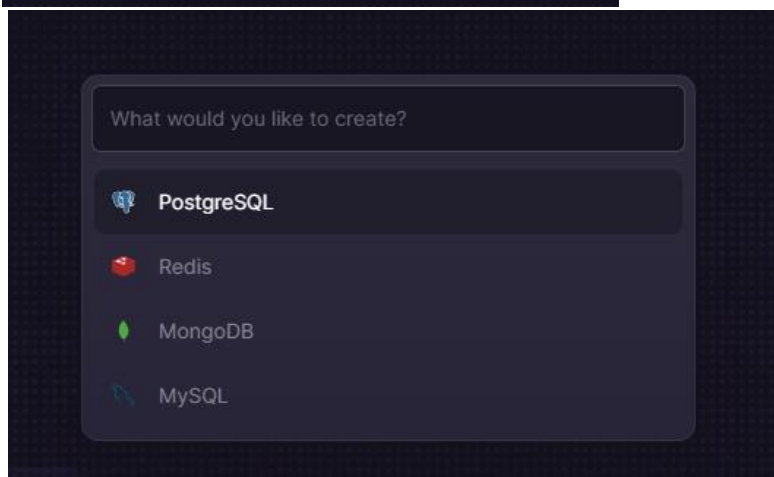
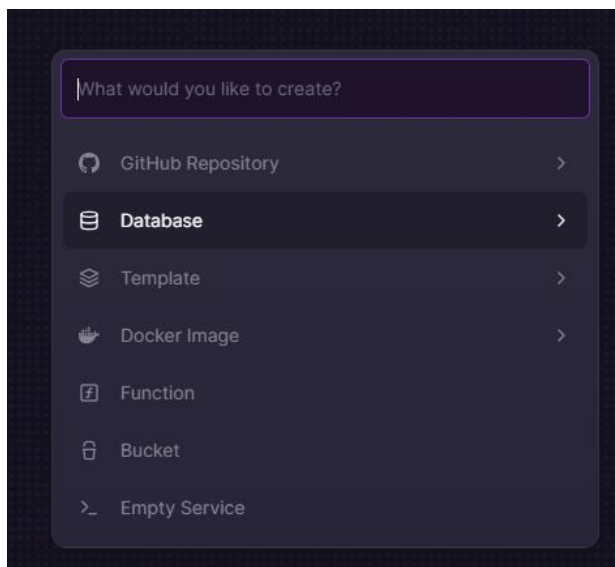
Deployment & Access Instructions

Accessing the Application:

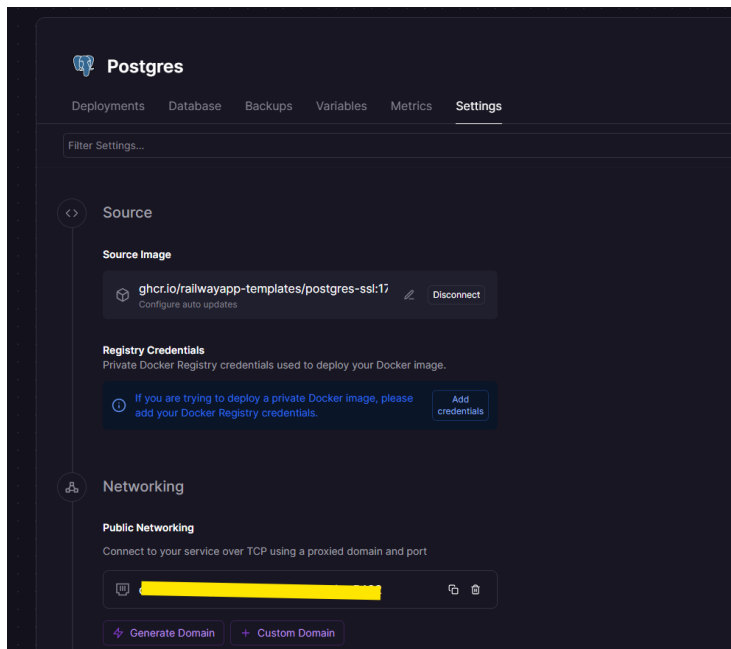
- **URL:** <https://project-management-django-production.up.railway.app/>

Database Deployment

1. Go to railway.com – create account and create new project. (can create account with github SSO)
2. The select database and select PostgreSQL DB



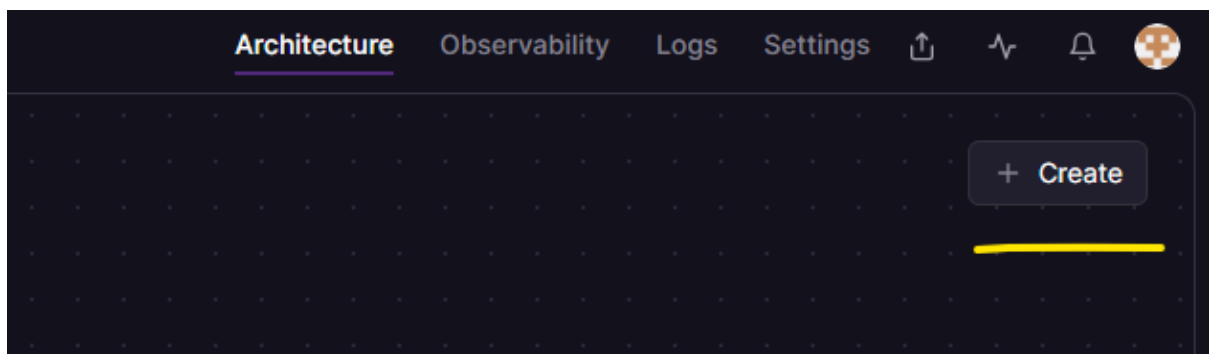
3. From the setting -> networking tab get the host, port username and password



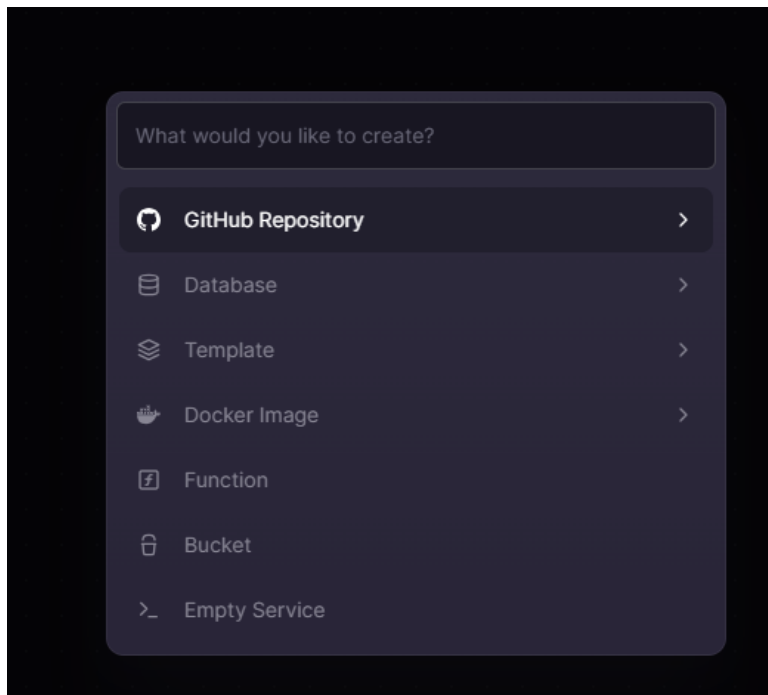
4. Connect from local pgadmin and run all your script to create schema, necessary tables and static data

Application Deployment - <https://railway.com/>

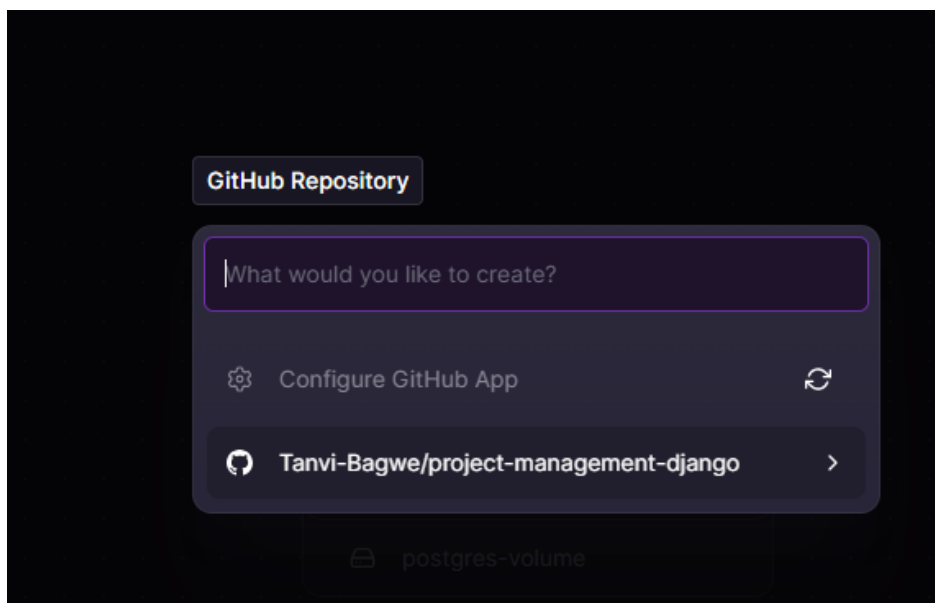
1. Before deploying to railway, you need to do following steps first
 - requirements.txt Railway needs to know what to install.
 - Add runtime.txt to project root to specify python version
 - Create a file named exactly Procfile (no extension) in your root folder.
 push all these above files to github.
2. Click on create from same project



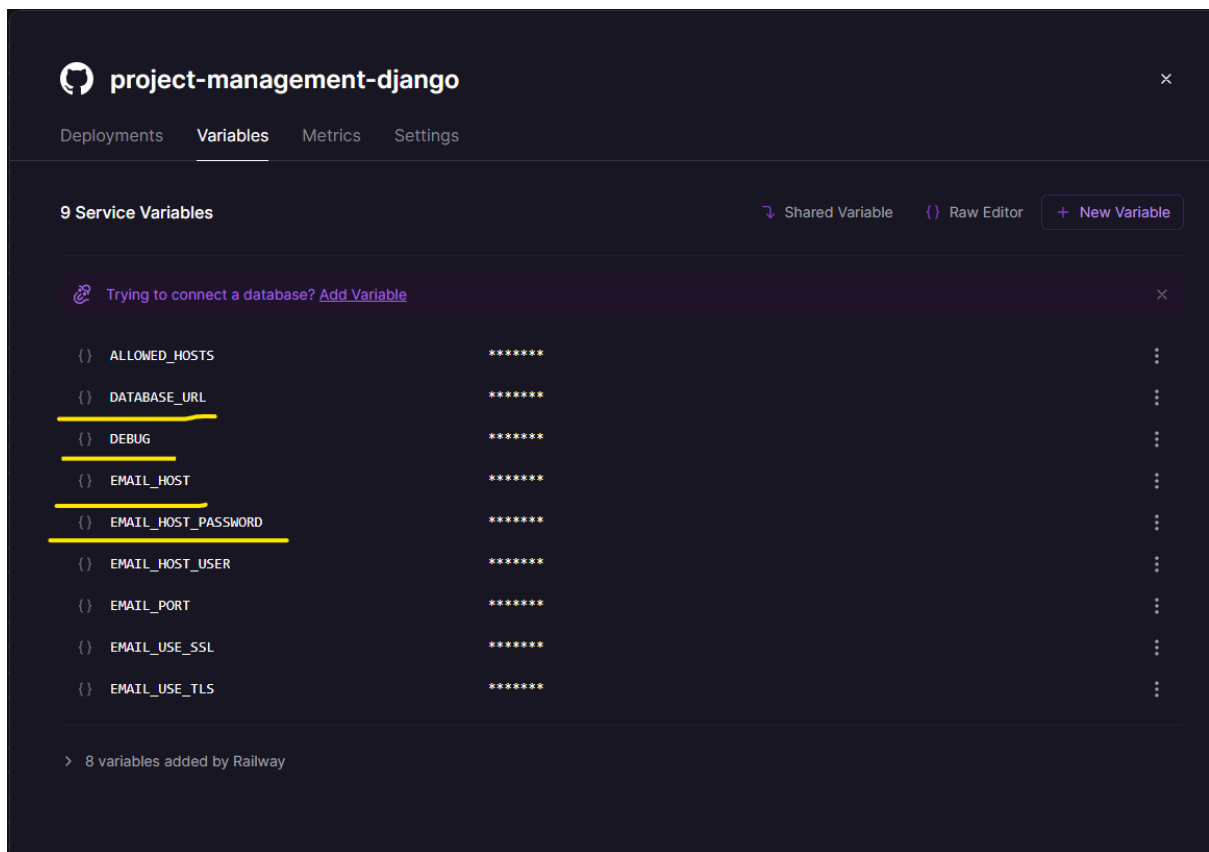
3. Select GitHub Repository



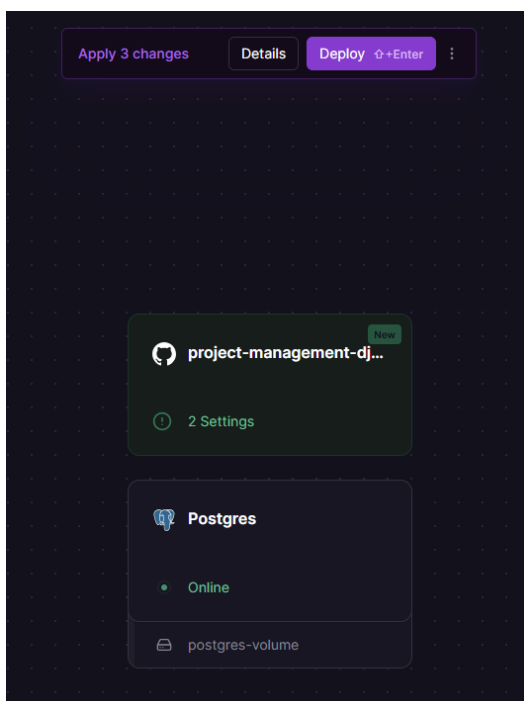
4. You should be able to see your public repos from github if you have connected the github account correctly



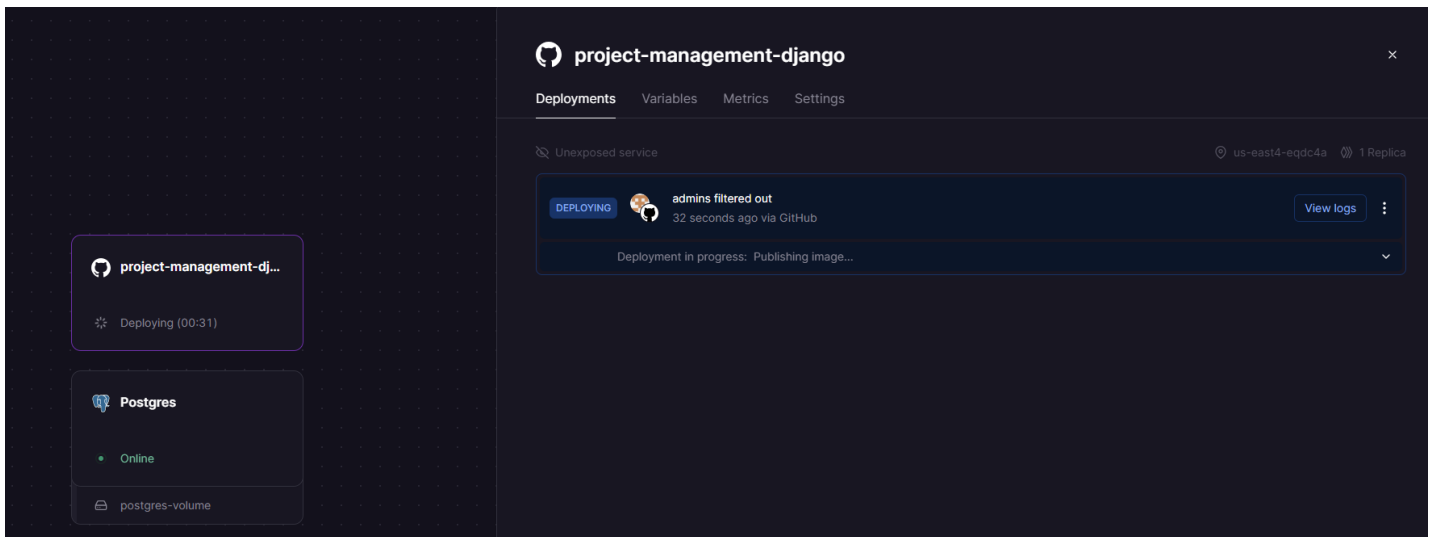
5. Go to variables of the service and add all the necessary variables like follow. (Highlighted ones are necessary)
DATABASE_URL: Automatically provided by Railway PostgreSQL.



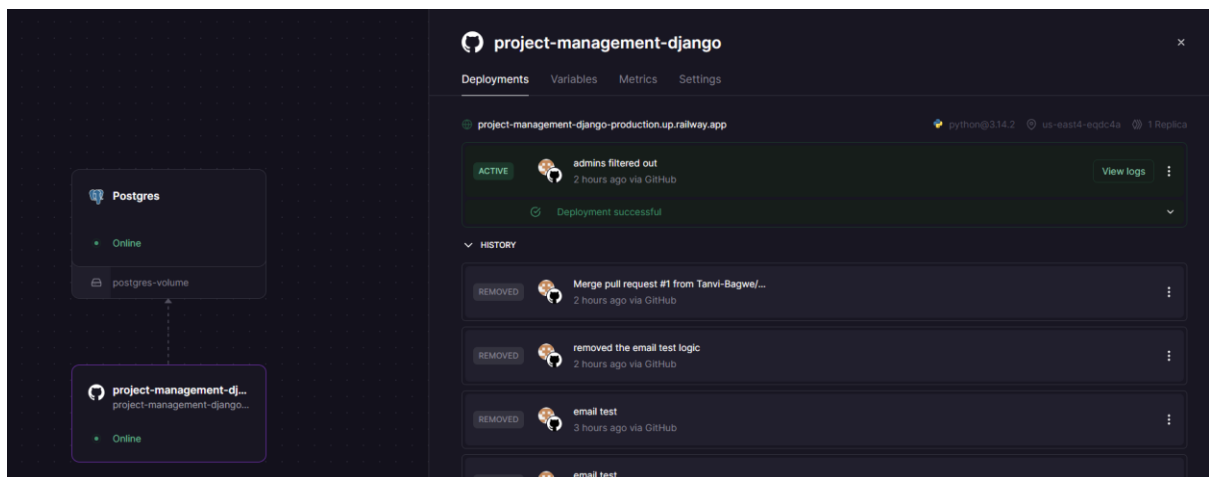
- When the variables are created and saved you should see the Deploy button, click on that to start the build and deploy process



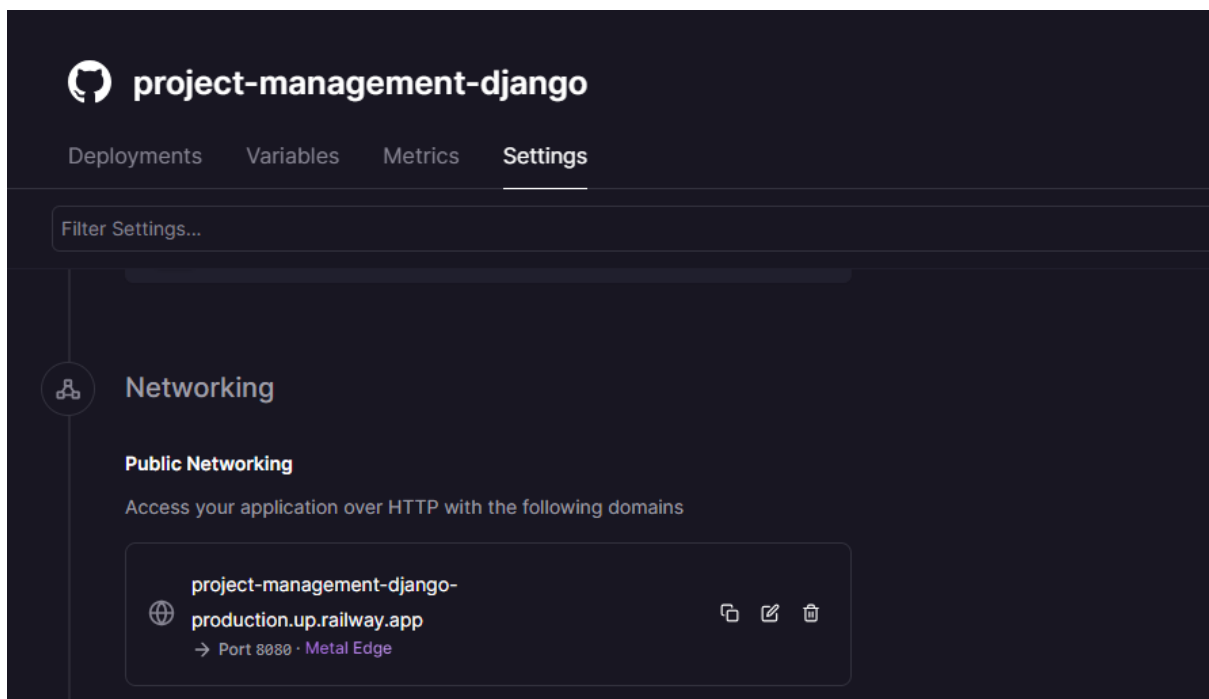
- You should see deploying status like this



8. Once the deployment is successful it should look like this



9. Then go to setting tab and you will be able to see the public URL from where you can access the app. You might need to enable the public networking if disabled



Local Installation

1. Clone the repository.
2. Install dependencies: ``pip install -r requirements.txt``.
3. Set up environment variables for ``EMAIL_HOST_USER`` and ``EMAIL_HOST_PASSWORD``.
4. Update the database string in ``settings.py``
5. Run queries from ``project_management.sql``
6. Run ``python manage.py migrate``.
7. If running on localhost server you might need to comment line 154 & 156 in ``settings.py`` -
`SECURE_PROXY_SSL_HEADER &`
`SECURE_SSL_REDIRECT`
8. Start server: ``python manage.py runserver``.
9. Access app at: ``http://127.0.0.1:8000/``